analog_neuron

Simulation and validation report · single-input analog ReLU neuron · Silicluster v3 (analog project)

SKY130A · 1.8 V · 180 µm × 115 µm · report generated 2026-10-20 from the result files of the runs listed in §4

$$y = 2 \cdot max(0, x - V_k), \quad V_k = 0.326 \cdot VDD$$

1. Summary

Quantity	Result	Target	
MAC linearity (INL, knee + 0.1 V … 1.2 V)	≤ 0.70 % (post-layout, 45 PVT); ≤ 0.76 % at R ±20 %; ≤ 0.81 % (Monte Carlo)	≤ 5 %	**pass**
Weight accuracy, mismatch (200-run Monte Carlo)	σ = 1.47 % (mean slope 1.957)	σ < 2–3 %	**pass**
Weight (slope) over PVT	1.904 – 1.995	2	
Knee over PVT	0.525 – 0.644 V (= 0.326 · VDD; flat over temperature and process)	VDD ratio	
Peak supply power	37.6 – 78.0 µW; 98.1 µW at R −20 %	< 100 µW (design target)	**pass**
Bias current (incl. R ±20 %)	1.13 – 3.89 µA	1 – 5 µA	**pass**
Bias start-up	starts at every PVT point, ≤ 3.0 µs after a 1 µs ramp; settled current identical across ramps and brown-out (≤ 0.000 %)	always starts	**pass**
Loop phase margin (post-layout)	signal loop ≥ 62.5°, knee loop ≥ 85.8°	≥ 60°	**pass**
DRC (Magic, sky130A `drc(full)`)	0 violations	clean	**pass**
LVS (netgen, `sky130A_setup.tcl`)	circuits match uniquely, no property errors	match	**pass**
Area	180.000 µm × 115.000 µm	180 µm × 115 µm	**pass**

PVT = 5 process corners (tt, ff, ss, sf, fs) × −40 / 27 / 125 °C × 1.62 / 1.8 / 1.98 V = 45 points. "Post-layout" = devices and parasitic capacitance extracted from the final layout with Magic.

2. Interface

Pin	Direction	Layer / location (µm, origin lower-left)	Electrical
`X`	analog input	metal3, right edge, x = 179.4, y = 81.27	0.3 – 1.2 V; ≈ 300 kΩ to an internal ≈ 0.26 V node (draws ≤ ≈ 3.2 µA)
`Y`	analog output	metal3, right edge, x = 179.4, y = 82.07	0 – ≈ 1.2 V; unbuffered, ≈ 300 kΩ source impedance
`VDD`	supply	metal5 horizontal rail, y = 111.5 – 114.5	1.8 V nominal (verified 1.62 – 1.98 V)
`VSS`	ground	metal5 horizontal rail, y = 0.5 – 3.5	0 V

Power grid: metal4 vertical straps (10 µm pitch, alternating VDD/VSS) and metal5 horizontal rails, as required by §11 of the call. Port declaration: `analog_neuron.v`.

3. Circuit

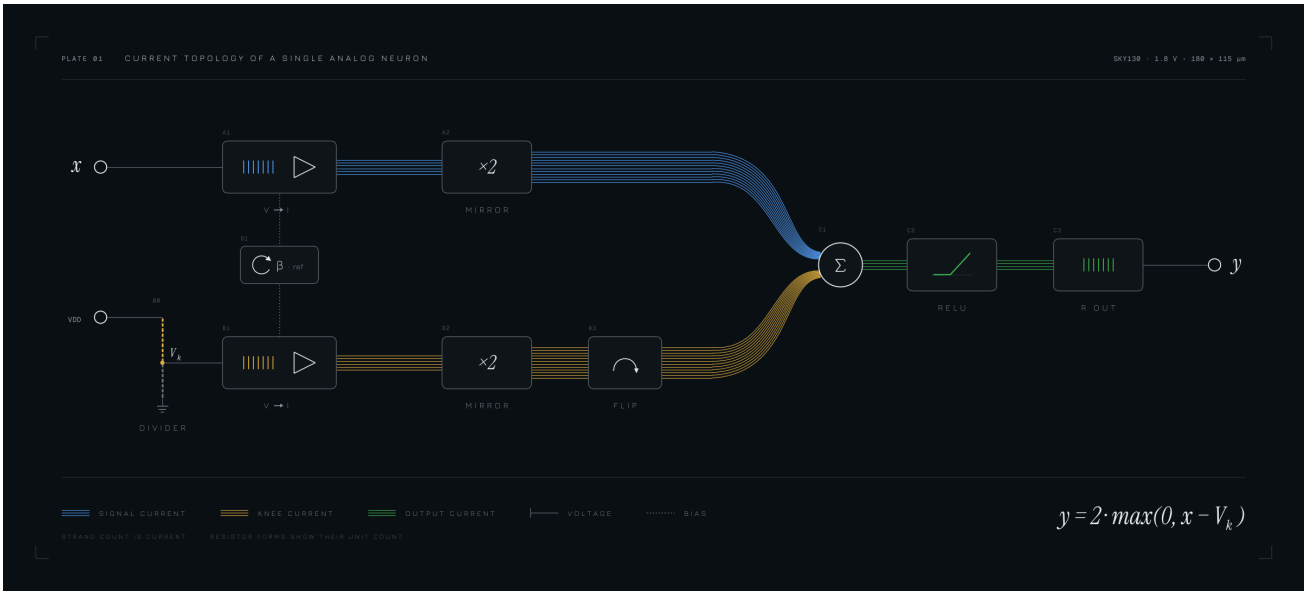


Figure 1 — Current topology. Strand count represents current: each ×2 mirror doubles its bundle, the knee current is subtracted at Σ and only the excess reaches the ReLU and R_{out}.

- **V→I converters.** An OTA with a PMOS input pair holds one end of R_{in} at an internal reference $V_{REF} \approx 0.142 \cdot VDD$, so $I_x = (x - V_{REF})/R_{in}$. A replica converter driven by the divider voltage V_k produces the knee current; V_{REF} cancels between the two.
- **Weight.** NMOS mirrors (W/L = 16/4, ×2 copies) set w = 2; the gain is $2 \cdot R_{out}/R_{in}$ with R_{out} and R_{in} built from matched xhigh_po unit resistors (7 units each, ≈ 43.4 kΩ/unit).
- **ReLU.** The knee current is flipped by a PMOS mirror and subtracted from the signal current; a PMOS diode can only supply the positive excess, which is copied into R_{out}.
- **Bias.** Beta-multiplier current reference with start-up (≈ 2 µA at 27 °C).
- **Compensation.** Miller capacitor 0.80 pF (MIM 20 × 20 µm) with a ≈ 72 kΩ nulling resistor per converter.

4. Simulation setup

Item	Value
Simulator / PDK	ngspice-47 (KLU); SKY130A via ciel, version 1689ac3f; models `sky130.lib.spice`
Corners	tt, ff, ss, sf, fs × −40 / 27 / 125 °C × 1.62 / 1.8 / 1.98 V; xhigh_po sheet resistance additionally at −20 % and +20 % (the PDK corners do not vary it and SkyWater lists its e-test specs as TBD)
Mismatch	200 Monte Carlo runs, corner `tt_mm` (device mismatch enabled)
Transfer test	power-up, then x ramped 0.29 → 1.21 V over 2 ms (quasi-static); output loaded only by stand-in pad ESD diodes
Stability test	Middlebrook double injection at each OTA output (cuts every feedback path), x = 0.3 / 0.75 / 1.2 V
Post-layout netlist	Magic extraction of the final layout: devices + 506 parasitic capacitors
Testbenches	`spice/neuron/neuron_tb.cir`, `neuron_pex_tb.cir`, `neuron_pex_mb_tb.cir`, `spice/bias/bias_tb.cir`, `bias_startup_tb.cir`, `bias_dcramp_tb.cir`; runner `scripts/sim.py`

5. Results

5.1 Transfer function

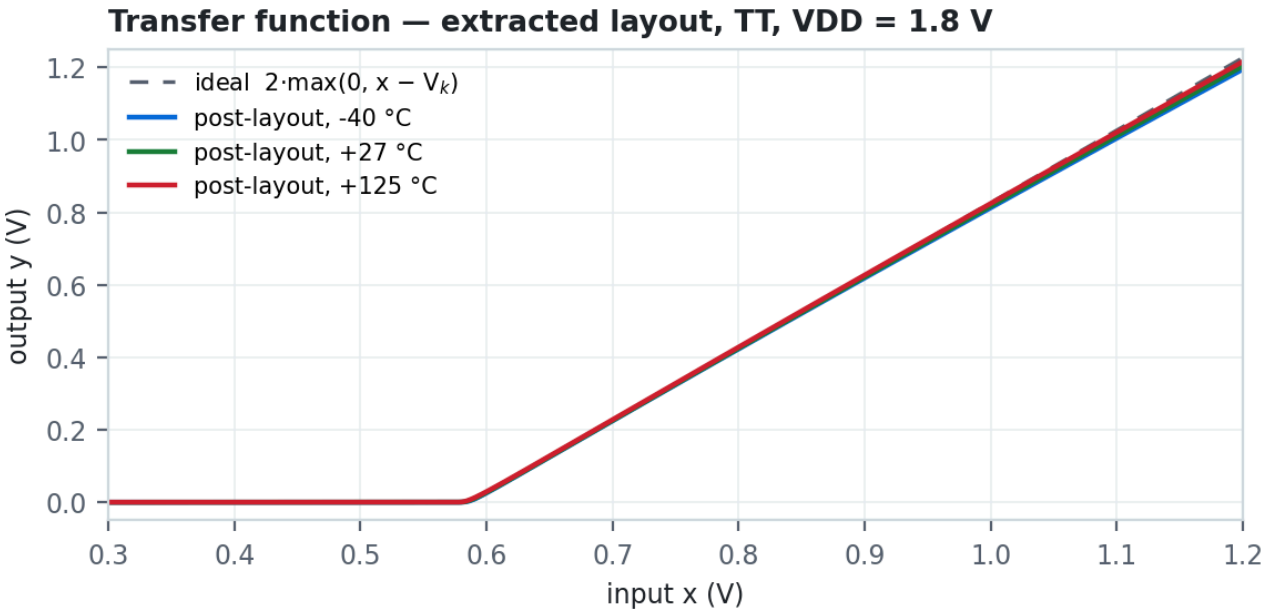


Figure 2 — Post-layout transfer at three temperatures against the ideal ReLU. The output is ≈ 0 below the knee and linear above it; temperature moves the curve by less than the line width.

5.2 All PVT points

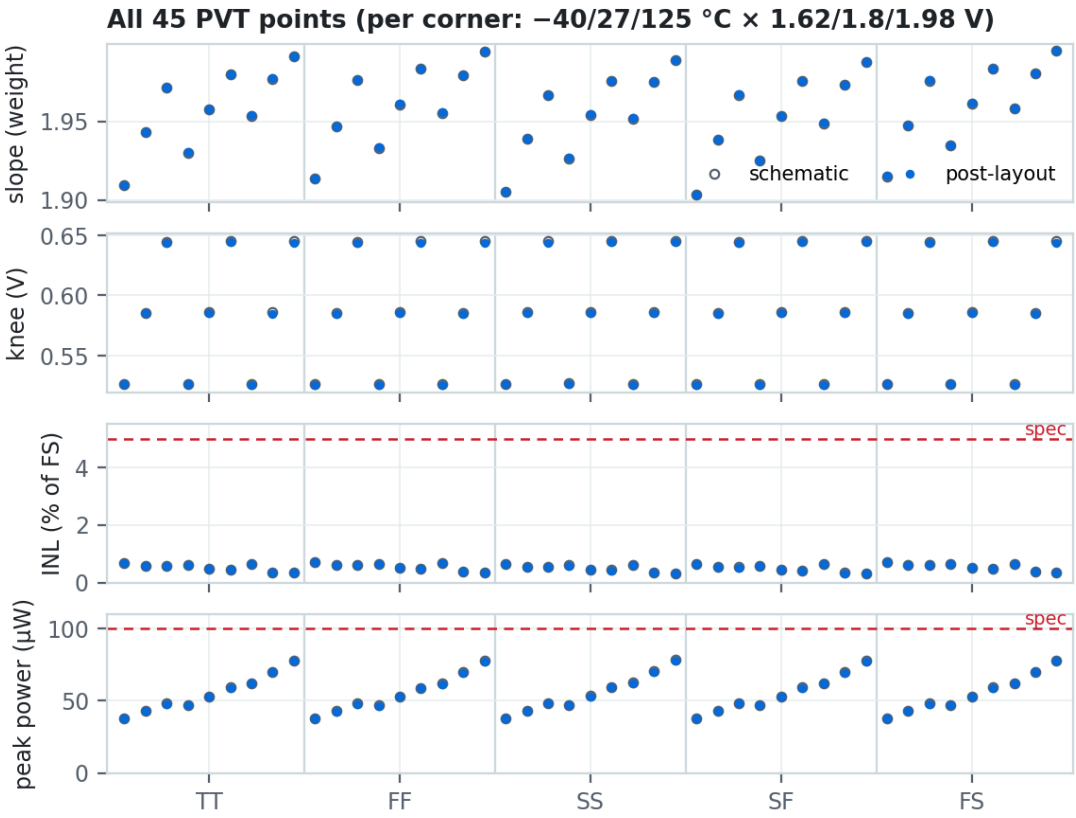


Figure 3 — Slope, knee, linearity and power at every PVT point, schematic (open) and post-layout (filled). The knee depends only on VDD (three levels per corner); layout changes the results by less than 0.1 %.

5.3 Mismatch

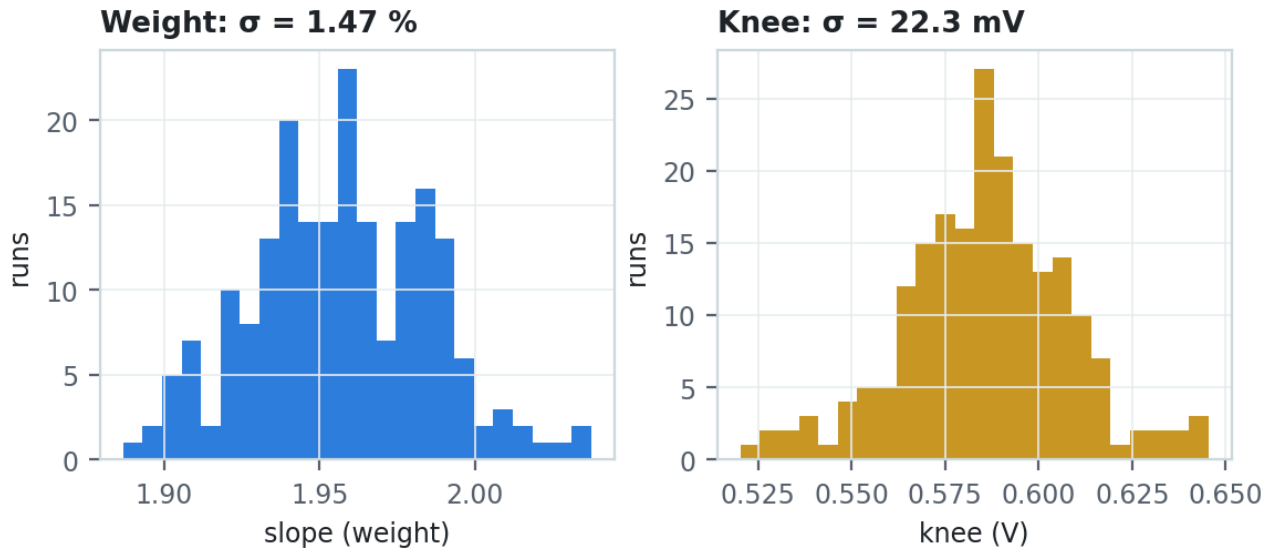


Figure 4 — 200-run Monte Carlo (tt_mm, 27 °C, 1.8 V). Weight σ 1.47 % meets the 2–3 % target; knee σ 22.3 mV (mean 0.586 V).

5.4 Bias reference and start-up

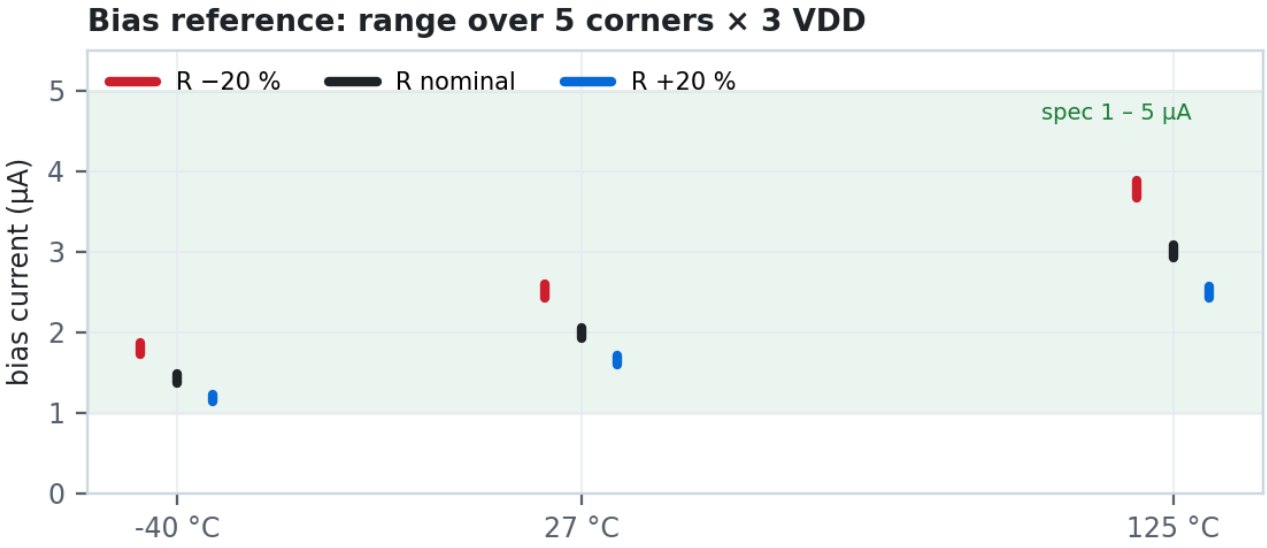


Figure 5 — Bias current range at each temperature for nominal and ±20 % resistance. Start-up was checked at all 45 points with a 1 µs ramp, a 100 µs ramp, a brown-out to 0.8 V and a quasi-static DC ramp; the reference turns on by VDD = 1.22 V at the latest.

5.5 Stability

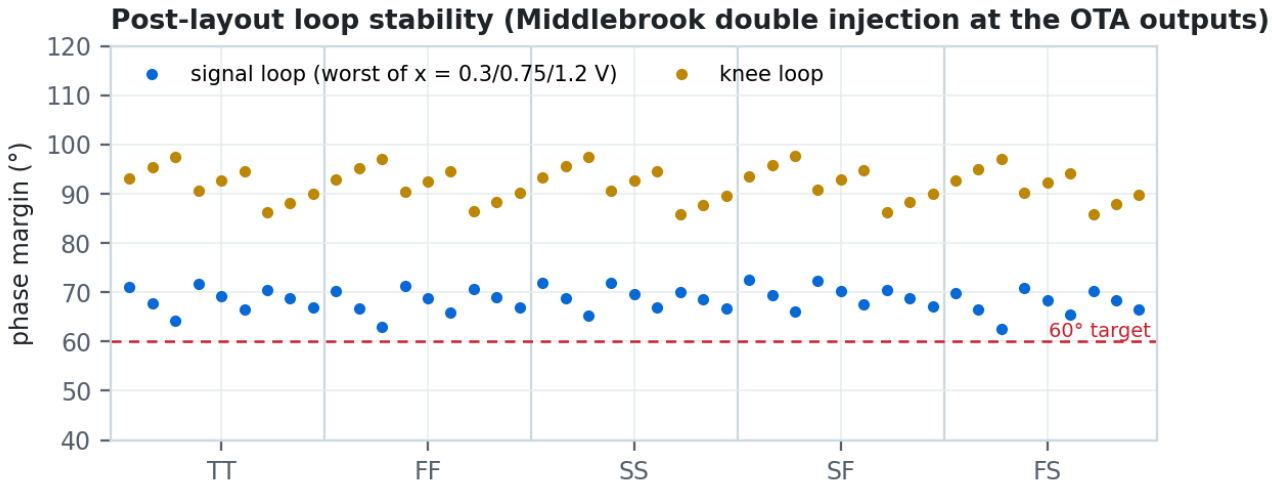


Figure 6 — Post-layout phase margin of both converter loops at all 45 PVT points. The knee loop has a second feedback path through the shared divider (its load moves V_{REF}); the measurement cuts at the OTA output so that path is included.

6. Physical verification

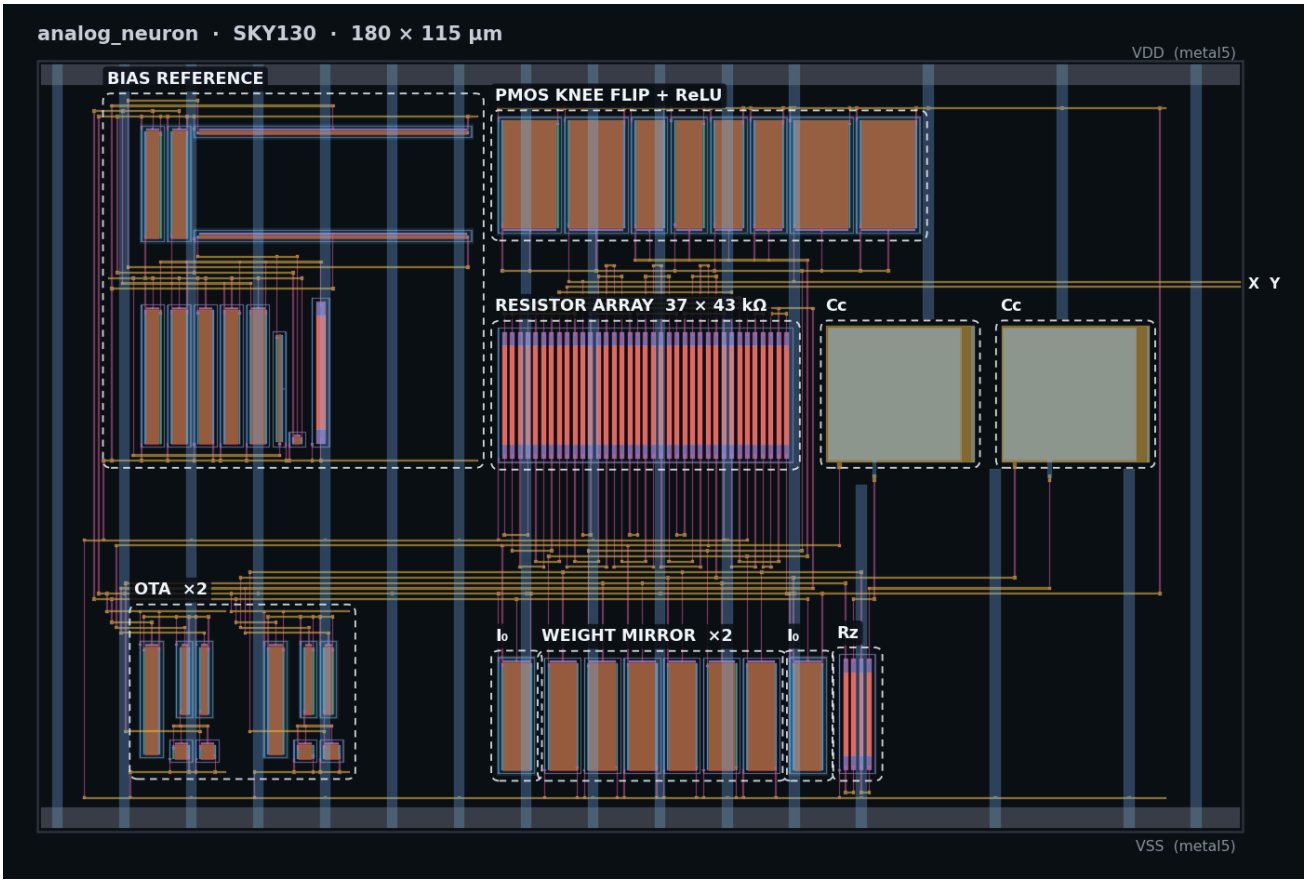


Figure 7 — Final layout, 180 µm × 115 µm, with functional blocks outlined. Power grid: metal4 vertical straps, metal5 horizontal rails.

DRC — Magic, sky130A `drc(full)`, top cell `analog_neuron`

```
No errors found.
DRC_COUNT 0
```

LVS — netgen with `sky130A_setup.tcl`, top cell against the reference netlist

```
Final result:
LVS_VERDICT PASS (topology and device properties match)
neuron_core:  Number of devices: 23                      |Number of devices: 23 | Number of nets:
16                        |Number of nets: 16
```

Each block (`ota5p`, `bias_ref`, `neuron_core`) also passes DRC and LVS on its own. Full logs are in `verification/` in this package. PASS is only declared when netgen reports a unique match *and* no property (device size) errors.

7. Reproducing the results

```
export PDK_ROOT=$HOME/.ciel PDK=sky130A
python3 scripts/sim.py spice/neuron/neuron_tb.cir --pvt --jobs 12          # schematic, 45 PVT
python3 scripts/sim.py spice/neuron/neuron_tb.cir --mc 200 --corner tt_mm --jobs 12
python3 scripts/sim.py spice/neuron/neuron_pex_tb.cir --pvt --jobs 12      # post-layout transfer
python3 scripts/sim.py spice/neuron/neuron_pex_mb_tb.cir --pvt --jobs 12   # post-layout stability
python3 scripts/sim.py spice/bias/bias_tb.cir --pvt                        # bias reference
```

8. Known limitations

- > **Unbuffered output.** Y has ≈ 300 kΩ source impedance. A 10 MΩ load reduces the gain by ≈ 2.8 %; a 1 MΩ load by ≈ 22 %. Linearity is unaffected. The output should be read by a high-impedance input.
- **xhigh_po spread.** The ±20 % sheet-resistance range is an assumption; at −20 % combined with 125 °C and 1.98 V the core reaches 98.1 µW.
- **Systematic weight.** The slope is 1–5 % below 2 (output-mirror drain-voltage difference), largest at low VDD and −40 °C.
- **Not run:** antenna-rule check.